FOUNDATION FOR
FOOD & AGRICULTURE
RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA18-SS-0000000005
Award Program	Sustainable American Aquaculture
Project Title	Development of environmental conditioning practices to decrease impacts of climate change on shellfish aquaculture
Grant Start Date	01/01/2018
Grant End Date	06/30/22
Total Grant Budget	$1,754,067.19
Total FFAR Budget	$878,956.07
Matching Funder(s)	University of Washington University of Rhode Island Jamestown S'Klallam Tribe Baywater Inc

Final Report Date	09/20/2022
Project Director/Principal Investigator Information	
Full Name	Steven Roberts
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Grantee Organization Information	
Organization Name	University of Washington
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Tax ID	91-6001537
Authorized Signing Official (ASO) Information	
Name	Robin Fransen
ASO Title	Program Coordinator, Office of Sponsored Programs
Email	rfrans@uw.edu
Phone	206.543.4043

General Information

1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Brinnon, Washington; Washington's 6th congressional district

2. How many new jobs were created because of this grant funding?

5

3. How many jobs were maintained because of this grant funding?

   3

4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

   No

5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

   No

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone 'and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

This project focused on improving aquaculture of geoducks, among the largest and most valuable burrowing clams. A critical production bottleneck is the hatchery phase. The primary goal of the project was to develop and assess hatchery performance enhancement methods through the use of environmental preconditioning, or "hardening." Trials were conducted on geoduck during early developmental stages to determine to what degree environmental preconditioning with low pH conditions improves performance. Through our efforts we determined hypercapnic conditions during post-larval development improved physiological performance and cellular stress response. In other words, moderate stress during early development can improve performance on subsequent exposure. In a separate study on juvenile geoducks we found exposure to low pH resulted in an increase in size and metabolic rate after 5 months under ambient conditions. The second objective of this work was to identify the underlying genomic mechanisms involved in enhanced performance. As part of the study looking at low pH exposure during post-larval development we were able to reveal transcriptional frontloading is responsible for improved responses to subsequent exposure. Pre-exposed clams frontloaded genes related to cellular quality control and immune response, resulting in the clams being better prepared for stressful conditions. Furthermore,

we demonstrated an underlying alteration in DNA methylation landscape coincides with beneficial phenotypic carry-over effects in geoduck clams. In addition to accomplishing the primary research objectives of this project, we also made additional contributions to the community including the production of a geoduck genome, development of new approaches in algae assessment and reproductive maturation assessment, and protocols for improved hatchery practices. In summary, this combined effort will contribute to the continued growth of shellfish aquaculture in the United States by offering practices and genomic resources to produce more resilient shellfish.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

There are two primary means by which the key findings can be used by the agriculture community. One is that they can start implementing moderate stress conditions during early development such that the length of time in the hatchery could be reduced (ie fast growth) or in the event clams were to experience stressful conditions later in production, they would perform better. This could be using decreased pH as was done as part of this study or a different stressor might be selected in areas where there are different environmental condition concerns (eg salinity, disease).

A second means by which the community can use the key findings is by the incorporation of genomic data into broodstock selection programs. Here we found that frontloading, or the constitutive expression of key genes, is responsible for increased resilience under environmental stress. Efforts to identify cohorts or genotypes that have these transcriptomic profiles could be a tool to complement conventional breeding programs. Likewise, epigenetic profiles could be targeted for broodstock selection or the community could use epigenetic profiles to optimize environmental manipulation scenarios.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S.

agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

The research outcomes of this effort will include the potential outcome of increased shellfish production under conditions where environmental stressors threaten resilience. Intermediate outcomes include the incorporation of our research products including geoduck genome, DNA methylation characterization, and related genomic resources into new research directions that could lead to improved outcomes.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The specific objectives of the project were to
A) Identify key stages in the geoduck life cycle when environmental conditioning can be applied for optimal benefits to productivity, and
B) Identify underlying mechanisms involved in enhanced performance.

There were no specific target dates.

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Major activities of the project include; [01] post-larval exposure experiment, [02] juvenile experiment with a short (10 day) interval prior to secondary exposure, [03] juvenile experiment with a long (100 day) interval prior to secondary exposure, [04] broodstock exposure experiment and offspring assessment, and [05] genome assembly and annotation.

The post-larval exposure experiment (see schematic below) set out to evaluate life-stage and stress-intensity dependence on tolerance to subsequent exposures (doi:10.1242/jeb.233932) and the functional role of transcriptomic profiles (doi:10.1111/mec.16644). We onditioned pediveliger larvae to ambient and moderately elevated pCO_2 for 110 days. Then, clams were exposed to ambient, moderate or severely elevated pCO_2 (750, 2800 or 4900 µatm, respectively) for 7 days and, following 7 days in ambient conditions, a 7-day third exposure to ambient or moderate pCO_2. Initial conditioning to moderate pCO_2 stress followed by second and third exposure to severe and moderate pCO_2 stress increased respiration rate, organic biomass and shell size, suggesting a stress-intensity-dependent effect on energetics. Additionally, stress-acclimated clams had lower antioxidant capacity supporting the hypothesis that stress during development affects oxidative status later in life. As part of this same study gene expression profiles were assessed. Pre-exposed geoducks frontloaded genes for stress and apoptosis/innate immune response, homeostatic processes, protein degradation and transcriptional modifiers. Pre-exposed geoducks were also responsive to subsequent encounters, with gene sets enriched for mitochondrial recycling and immune defense under elevated pCO_2 and energy metabolism and biosynthesis under ambient recovery. This transcriptomic data indicates that pCO_2 priming during post-larval periods enhances resilience via gene expression regulation.

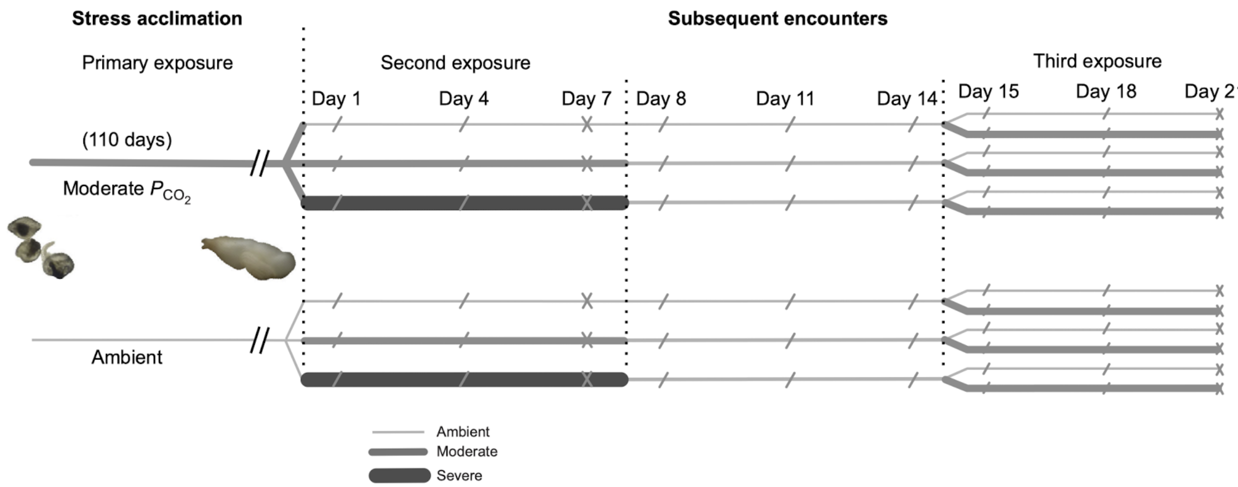


Fig. 1. Schematic of the experimental design. Line width represents the P_{CO_2} treatment during the primary exposure period and subsequent 7 day exposure periods (thin line, ambient P_{CO_2}; mid line, moderate elevated P_{CO_2}; wide line, severe P_{CO_2}). Single dashes '/' indicate sampling days for respiration and shell growth measurements and '×' indicates sampling days for both respiration and growth measurements and fixed tissues for subcellular analysis.

The juvenile experiment with the short interval (see schematic below) we tested the hypothesis that repeated stress exposure under elevated pCO2 can enhance intragenerational performance (doi:10.1093/conphys/coaa024). We measured the respiration rate and shell growth of juvenile geoduck in a commercial hatchery under repeated acute periods (~6–10 days) of elevated pCO2 and the longer term (~5 months) carryover effects. Juvenile geoduck exposed to low pH for 10 days recovered from metabolic depression under subsequent stress exposure and conditioned animals showed a significant increase in both shell length and metabolic rate after 5 months under ambient conditions. This demonstrates an environmental memory and compensatory growth as indicators of enhanced resilience from stress preconditioning.

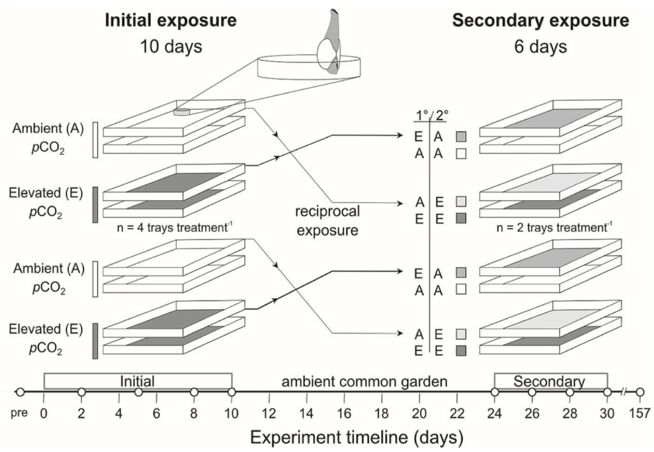
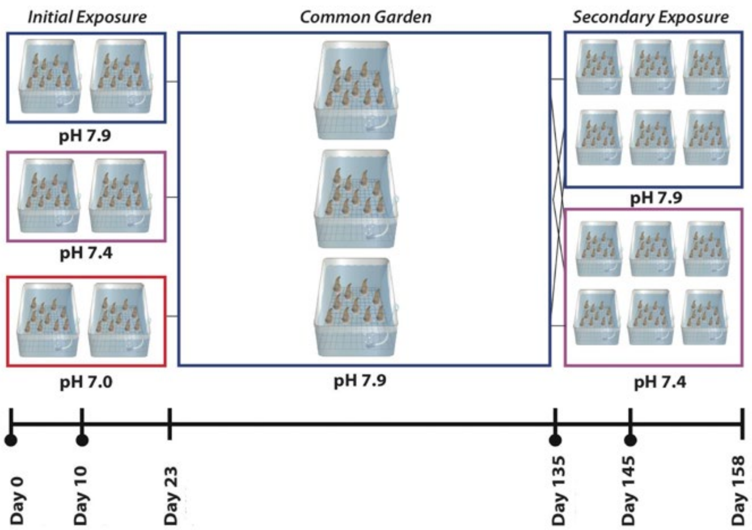


Figure 1: Schematic of the repeated exposure experimental design for two exposure trials, initial (10-day) and secondary (6-day), in ambient and elevated pCO_2 treatments. Timeline displays respiration and growth measurements as solid white circles

Another activity was the juvenile experiment with longer interval period (see schematic; doi:10.1101/2022.06.24.497506). Juveniles were initially exposed to one of three seawater pH conditions, followed by ambient common-garden conditions, then a second reciprocal exposure to ambient pH or pH 7.4. Within 10 days of the initial low pH exposure, juvenile clams showed decreased shell size relative to ambient pH with corresponding differential DNA methylation. Following four months of ambient common-garden conditions, juveniles initially exposed to low pH compensatorily grew larger, with DNA methylation indicative of these phenotypic differences, demonstrating epigenetic carryover effects persisted months after initial exposure. Functional enrichment analysis of differentially methylated genes revealed regulation of signal transduction through widespread changes in the Wnt signaling pathways that influence cell growth, proliferation, tissue and skeletal formation, and cytoskeletal change. After 10 days of secondary exposure to pH 7.4, naive juvenile clams were more sensitive to low pH compared to those initially exposed, showing reduced growth and having nearly a 2-fold greater change in DNA methylation. This work demonstrated an underlying alteration in the DNA methylation landscape that coincides with a beneficial phenotypic carry-over effect. These epigenetic findings provide an exploitable role in environmental manipulation that would benefit aquaculture.



Another major activity was examining the effect of broodstock conditioning on offspring performance. To investigate whether or not stress conditioning might improve performance in aquaculture settings, we explored the effects of variable seawater pCO2 conditioning on adults and their offspring. While variable high pCO2 conditioning had no effect on broodstock survival, it had a negative effect on juvenile offspring survival with poorer survival in offspring from preconditioned parents. However juvenile growth was increased in exposed offspring from preconditioned parents despite a growth delay they exhibited in the late larval stage. Moreover, exposed offspring from preconditioned parents were able to maintain a lower and less variable metabolic rate throughout all stages of early development than offspring from naive parents. Taken together, short-term, variable stress parental preconditioning can give rise to beneficial phenotypes in offspring (achieving greater size with less metabolic activity) and could be adopted in industry practices for shellfish rearing.

The final and fifth major activity of this project was the annotation of the genome (doi:10.1101/2022.06.24.497506). The assembly resulted in a set of 18 chromosome- scale scaffolds containing 1.42 Gbp of sequence. Juicebox correction resulted in a final set of 18 scaffolds spanning 942 Mbp, with a scaffold N50 of 57,743,597 bp and a scaffold N90 of 34,923,512 bp. Genome annotation identified 34,947 genes and 236,960 coding sequence regions, corresponding to 38,326 mRNA features. This resource was critical for our own genomic work and will also provide important resources for other researchers studying this and related species.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

This was primarily done through careful planning with team members and outside collaborators and all work has either already gone through or will soon go through a rigorous peer review process. In addition, we provide data and code for all efforts so this too can be fully evaluated by the research community.

2.8. List key stakeholders that could be served by results of this project.

Shellfish Aquaculture Industry

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Primary means of dissemination of information have included presentations, online lab notebooks, and a website dedicated to sharing information on the project (https://safsoa.wordpress.com). A list of presentations is provided

NACE - 2019 Northeast Aquaculture Conference and Exposition. Gurr et al. Effects of repeated short-term exposure to ocean acidification on juvenile Pacific geoduck *Panopea generosa*

Gurr, S.J., Vadopalas, B., Roberts, S.R., Putnam, H.M., September 2019. Metabolic recovery and compensatory shell growth of juvenile Pacific geoduck *Panopea generosa* following short-term exposure to acidified seawater. Pacific Coast Shellfish Growers Association 73rd Annual Shellfish Conference and Tradeshow. Portland, Oregon.

Gurr, S. J., Trigg, S. A., Vadopalas, B., Roberts, S. B., & Putnam, H. M. January 2021. 'Environmental learning 'in a tolerant commercial clam: insights from phenotypic and subcellular adjustments to hypercapnic seawater. Society for Integrative and Comparative Biology. Virtual conference.

Gurr, S. J., Trigg, S. A., Vadopalas, B., Roberts, S. B., & Putnam, H. M. 2020. Repeat exposure to hypercapnic seawater modifies performance and oxidative status in a tolerant burrowing clam. *bioRxiv*.

Gurr, S.J., Vadopalas, B., Roberts, S.R., Putnam, H.M., February 2020. Thresholds of intragenerational pCO2 conditioning on metabolism, oxidative stress resilience, and DNA methylation of juvenile Pacific geoduck *Panopea generosa*. Ocean Sciences Meeting 2020. San Diego, CA.

Trigg, S.A., Putnam, H.M., Gurr, S.J., Roberts, S.B., Mitchell, K.R., Vadopalas, B., Henderson, M. January 2021. Exploring the tolerance of Pacific geoduck to low pH through comparative physiology, genomics, and DNA methylation. Society for Integrative and Comparative Biology. Virtual conference.

Trigg, S.A. January 2020. Influence of ocean acidification on DNA methylation patterns in geoduck. The Plant and Animal Genome XXVII Conference (PAG) Town and Country San Diego, CA.

Trigg, S.A. and Vadopalas, B. March 2020. Salish Sea, Ocean Acidification & Geoducks, Oh My! Salish Sea Expeditions Training workshop. Northwest Maritime Center, Port Townsend, WA.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

nothing to report

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to

500-word limit)

This project has supported the training of a graduate student and post-doc. Individual development plans were used to support professional development. The trainees complete a self assessment and then the trainee and advisor plan the milestones and timelines for the following: 1 month, 1 semester, 1 year, Dissertation, Career. We use a SMART approach to Goals and Milestone setting: Specific, Measurable, Relevant, Timebound. These IDPs are revisited every 6 months.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Undergraduates 2
Graduate Students 1
Post-docs 1

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR

TECHNICAL REPORTS

SCHOLARLY PUBLICATIONS

3.2. Please provide a list of citations for the information products produced during the project.

Gurr SJ, Rollando C, Chan LL, Vadopalas B, Putnam HM, Roberts SB (2018) Alternative image-based technique for phytoplankton cell counts in shellfish aquaculture (no. 1001481). Nexcelom.

Gurrr SJ. Geoduck Hatchery Manual DOI: 10.5281/zenodo.7086401

Samuel J. Gurr, Brent Vadopalas, Steven B. Roberts, Hollie M. Putnam (2020) Metabolic recovery and compensatory shell growth of juvenile Pacific geoduck Panopea generosa following short-term exposure to acidified seawater Conservation Physiology, Volume 8, Issue 1, coaa024. doi:10.1093/conphys/coaa024

Samuel J. Gurr, Shelly A. Trigg, Brent Vadopalas, Steven B. Roberts, Hollie M. Putnam (2021) Repeat exposure to hypercapnic seawater modifies growth and oxidative status in a tolerant burrowing clam J Exp Biol 1 July 2021; 224 (13) doi:10.1242/jeb.233932

Samuel J. Gurr, Shelly A. Trigg, Brent Vadopalas, Steven B. Roberts, Hollie M. Putnam (2022) Acclimatory gene expression of primed clams enhances robustness to elevated pCO2 Molecular Ecology. doi:10.1111/mec.16644

Hollie M. Putnam, Shelly A. Trigg, Samuel J. White, Laura H. Spencer, Brent Vadopalas, Aparna Natarajan, Jonathan Hetzel, Erich Jaeger, Jonathan Soohoo, Cristian Gallardo-Escárate, Frederick W. Goetz, Steven B. Roberts (2022) Dynamic DNA methylation contributes to carryover effects and beneficial acclimatization in geoduck clams. bioRxiv https://doi.org/10.1101/2022.06.24.497506

Byrne, M, Foo SA, Ross, PM Putnam, HM. (2020) Limitations of cross- and multigenerational plasticity for marine invertebrates faced with global climate change. Global Change Biology https://doi.org/10.1111/gcb.14882

Shelly A. Trigg, Samuel Gurr, Kaitlyn R. Mitchell, Brent Vadopalas, Hollie M. Putnam, Steven B. Roberts. Integrated analysis of gonad maturation in geoduck. (*in prep*)

Shelly A. Trigg, Samuel J. Gurr, Kaitlyn R. Mitchell, Brent Vadopalas, Hollie M. Putnam, and Steven B. Roberts. Effect of broodstock conditioning on offspring development and stress response in Pacific geoduck. (*in prep*).

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Samuel J. Gurr, Brent Vadopalas, Steven B. Roberts, Hollie M. Putnam (2020) Metabolic recovery and compensatory shell growth of juvenile Pacific geoduck Panopea generosa following short-term exposure to acidified seawater Conservation Physiology, Volume 8, Issue 1, coaa024. doi:10.1093/conphys/coaa024

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Samuel J. Gurr, Shelly A. Trigg, Brent Vadopalas, Steven B. Roberts, Hollie M. Putnam (2021) Repeat exposure to hypercapnic seawater modifies growth and oxidative status in a tolerant burrowing clam J Exp Biol 1 July 2021; 224 (13) doi:10.1242/jeb.233932

Samuel J. Gurr, Shelly A. Trigg, Brent Vadopalas, Steven B. Roberts, Hollie M. Putnam (2022) Acclimatory gene expression of primed clams enhances robustness to elevated pCO2 Molecular Ecology. doi:10.1111/mec.16644

Hollie M. Putnam, Shelly A. Trigg, Samuel J. White, Laura H. Spencer, Brent Vadopalas, Aparna Natarajan, Jonathan Hetzel, Erich Jaeger, Jonathan Soohoo, Cristian Gallardo-Escárate, Frederick W. Goetz, Steven B. Roberts (2022) Dynamic DNA methylation contributes to carryover effects and beneficial acclimatization in geoduck clams. bioRxiv https://doi.org/10.1101/2022.06.24.497506

Byrne, M, Foo SA, Ross, PM Putnam, HM. (2020) Limitations of cross- and multigenerational plasticity for marine invertebrates faced with global climate change. Global Change Biology https://doi.org/10.1111/gcb.14882

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

https://safsoa.wordpress.com/ - general website sharing research progress

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No

	# Filed (Enter Numeric	# Approved (Enter Numeric Value)	Patent Numbers (Separat

	Value)		ed by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

We provide links to repositories in manuscripts, discuss products in presentations, and share on social media. We ensure others can access and use be providing code and data together in repositories with documentation.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

[DATA SET 01] Samuel Gurr. (2019). Metabolic recovery and compensatory shell growth of juvenile Pacific geoduck Panopea generosa following short-term exposure to acidified seawater. Zenodo. https://doi.org/10.5281/zenodo.3588326

[DATA SET 02] Samuel Gurr. (2020). Repeat exposure to hypercapnic seawater modifies performance and oxidative status in a tolerant burrowing clam. Zenodo. https://doi.org/10.5281/zenodo.3903019

[DATA SET 03] "Repeat exposure to hypercapnic seawater modifies performance and oxidative status in a tolerant burrowing clam" Raw sequence reads are deposited in the SRA (Accession: PRJNA740307; BioProject: Transcriptome profiles of Panopea generosa under hypercapnic seawater) https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA740307

[DATA SET 04] Samuel J. Gurr (2022). Acclimatory gene expression of primed clams enhances robustness to elevated pCO2. https://doi.org/10.5281/zenodo.6908630

[DATA SET 05] "Dynamic DNA methylation contributes to carryover effects and beneficial acclimatization in geoduck clams" The genomic DNA, transcriptomic, and DNA bisulfite sequence data generated in this study have been submitted to the NCBI BioProject database (https://www.ncbi.nlm.nih.gov/bioproject/) under accession numbers PRJNA316601; PRJNA529226 and PRJNA646071; and PRJNA566166, respectively.

[DATA SET 06] Dynamic DNA methylation contributes to carryover effects and beneficial acclimatization in geoduck clams. OSF https://doi.org/10.17605/OSF.IO/YEM8N

[DATA SET 07] Integrated analysis of gonad maturation in geoduck. https://github.com/shellytrigg/paper-GeoduckReproDev_pH

[DATA SET 08] Effect of broodstock conditioning on offspring development and stress response in Pacific geoduck. https://github.com/shellytrigg/paper-GeoduckTransgen_var.pH

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Data sets 02-04 are from the same experiment examining environmental influences on post-larval to early juvenile geoducks.
Data sets 05 and 06 represent data from an experiment designed to characterize epigenetic mechanisms associated with environmental memory in geoducks

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

[DATA SET 01]
Author: Samuel Gurr.
Year: 2019
Title: Metabolic recovery and compensatory shell growth of juvenile Pacific geoduck Panopea generosa following short-term exposure to acidified seawater.
Repository: Zenodo
url: https://doi.org/10.5281/zenodo.3588326
Number of files: 15
Filetype: csv
Total size: 820 kB
License: CC attribution 4.0
DOI: 10.5281/zenodo.3588326

[DATA SET 02]
Author: Samuel Gurr
Year: 2020
Title: Repeat exposure to hypercapnic seawater modifies performance and oxidative status in a tolerant burrowing clam. Z
Repository: Zenodo
url: https://doi.org/10.5281/zenodo.3903019
Number of files: 201
Filetype: csv, text, pdf
Total size: 3.9 MB
License: CC attribution 4.0
DOI: 10.5281/zenodo.3903019

[DATA SET 03]
Provided by NCBI https://d.pr/f/BmN3B4

[DATA SET 04]
Author: Sam Gurr
Year: 2022
Title: Acclimatory gene expression of primed clams enhances robustness to elevated pCO2.
Repository: Zenodo
url: https://doi.org/10.5281/zenodo.6908630
Number of files: 201
Filetype: csv, pdf, text

Total size: 179 MB
License: CC attribution 4.0
DOI: 10.5281/zenodo.6908630

[DATA SET 05]
Provided by NCBI https://d.pr/f/08datO

[DATA SET 06]
Authors: Steven Roberts, Sam White, Shelly A. Trigg, Hollie Putnam
Year: 2019
Title: Panopea_generosa_genome
Repository: Open Science Framework
url: https://osf.io/yem8n/
Number of files: 100+
Filetype: csv, pdf, text
Total size: 21 GB
License: CC attribution 4.0
DOI: 10.17605/OSF.IO/YEM8N

[DATA SET 07]
Author: Shelly Trigg
Year 2020
Title: paper-GeoduckReproDev_pH
Repository: GitHub
Extended title: Integrated analysis of gonad maturation in geoduck.
url: https://github.com/shellytrigg/paper-GeoduckReproDev_pH
Filetype: csv, pdf, text

[DATA SET 08]
Authors: Sam Gurr, Shelly Trigg
Year: 2020
Title: paper-GeoduckTransgen_var.pH
Repository: GitHub
Extended title: Effect of broodstock conditioning on offspring development and stress response in Pacific geoduck.
url: https://github.com/shellytrigg/paper-GeoduckTransgen_var.pH
Filetype: csv, pdf, text

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Steven Roberts

PI Signature: __ Steven Roberts __________________ Date: _9/20/22________________

Authorized Signing Official (ASO) Name: ________Robin Fransen_______

ASO Signature: __Robin Fransen 2022.09.22 16:19:14 -07'00'__________ Date: _______9/22/2022_________